

Muhammad Essa

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EDUCATION

Dakota State University

Bachelor of Science in Cyber Operations, Minor in Network Security

Madison, SD

Aug. 2024 – May 2028

EXPERIENCE

Cybersecurity Competitor & CTF Participant

June 2023 – Present

Dakota State/ Self

Madison, SD

- * **UND Cyber Hawks Qualifier** Ranked 6th out of 101; qualified
- * **CyberBay CTF** Placed 13th out of 200 teams
- * Placed 3rd in NCAE Cyber Games, competing in network security, cryptography, and digital forensics
- * Competed in CyberCup (SimSpace), focusing on real-world cybersecurity scenarios
- * Engaged in HTB Binary Badlands, Dakota Con CTF, and DSU vs. WGU CTF, Cyber Summit Minneapolis strengthening skills in binary exploitation, reverse engineering, and penetration testing.

Cyber Academy Enrichment Programming Leader

April. 2025 – Present

Dakota State

Madison, SD

- * Coordinate and lead enrichment programming for the Governor's Cyber Academy and K-12 Initiatives at DSU.
- * Design and deliver hands-on cyber modules and mini-CTFs for K-12 classrooms; support teachers with lesson delivery, device/VM setup, and troubleshooting
- * Mentor student cohorts and model professionalism, integrity, and safe computing practices; collect feedback to continually improve session quality.

GenCyber Camp Counselor

June 2025 – June 2025

Dakota State University

Madison, SD

- * Facilitated daily cybersecurity sessions (networking, Linux, web security, crypto) using hands-on labs and mini-CTFs.
- * Supported teachers with lesson delivery, lab/VM setup, troubleshooting, and classroom management.
- * Coordinated camp logistics and quick assessments; gathered feedback to refine activities and boost engagement.

Campus Watch Specialist

August 2025-Present

Dakota State

Madison, SD

- * Work in a team to make Campus secure. Make rounds and assist people in terms of security inside college

PROJECTS

Velocity | C++

September 2025 – Present

- * Built a C++ vehicle-simulation playground to practice OOP, data structures, and algorithms; roadmap includes maps, races, and garages.
- * Structured the codebase into **Core**, **App**, and **Tests** modules with a Visual Studio solution/project for clean builds and separation of concerns

GameHub | C

May 2024 – May 2024

- * Developed in C, Game Hub offers two classic games: Tic-Tac-Toe and Number Guessing Game.
- * Implemented game logic for Tic-Tac-Toe, including input validation, win detection, and user feedback
- * Demonstrated core programming skills in loops, conditionals, arrays, and functions, with a modular design.
- * Focused on user-friendly design with a looped menu structure, enabling multiple rounds of play without restarting.

TECHNICAL SKILLS

Languages: JS, Python, C/C++, SQL , HTML/CSS, R

Developer Tools: Git, Docker, Google Cloud Platform, VS Code, Visual Studio, PyCharm

Libraries: pandas, NumPy, Matplotlib